

Patient Tracker System

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INFO-C 450: System Design

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Problem Statement:

A visit to the doctor's office or hospital is often met with stress from the start. It does not help that upon arrival, the patient is expected to fill out paperwork that feels excessive. There are many forms that patients fill out each time they enter a healthcare setting. There are still hospitals that use paper records instead of electronic file keeping, which can slow down the process of receiving results and make it more difficult for patients to access their health information.

A new patient tracker system should be electronic and keep all patient information and records in one easily accessible place. Patients should be able to log into the system to set up appointments, see test results, and access other relevant health information. This would speed up getting answers about health concerns and allow patients to stay involved in their own health. Patients would not have to wait for calls from providers and administration, the information would all be available to them immediately.

Glossary of Terms:

- Appointment Management- Schedule, reschedule, or cancel appointments
- Reminders- Notifications sent straight to a patient's phone or email address, reminding them to confirm appointments.
- Medical Record- A patient's medical history with past and current health issues. Test results and patient medications should also be listed here.
- Patient tracker- A portal for patients to see all their health information in one place.

System Requirements: •

Functional requirements:

No.	Priority Weight	Description
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REQ-1	High	Patient creates an account in the portal. Here they will be able to view all their medical information.
REQ-2	High	Patients can view, schedule, and cancel appointments to see providers from their portal account.
REQ-3	High	Patients can view their test results as soon as they are available.
REQ-4	High	Doctors and other medical staff can update records, so patients can see up to date treatment plans.
REQ-5	High	Patients can sign up to receive reminders of upcoming appointments and can
		reschedule or cancel if needed.

REQ-6	Medium	Patients can communicate directly with their providers and request prescription refills straight from the portal.
REQ-7	Medium	Administrative workers should be able to go into the system and make appointment changes.
REQ-8	Medium	Doctors and other medical professionals can use the portal to update availability and notify patients when they need to reschedule or cancel appointments.

- Nonfunctional Requirements:

No.	Priority Weight	Description
NFR-1	High	Functionality- The patient tracker system must allow patients, medical staff,
		administration, etc. to access and use the portal.

NFR-2	High	Usability- The patient tracker system should be selfexplanatory and easily accessible for all users regardless of age, disabilities, etc.
NFR-3	High	Reliability- The patient tracker must be able to support many users at one time. The information stored on the system should be secure.
NFR-4	High	Performance- Users should be able to access necessary information quickly.
NFR-5	Medium	Supportability- The system should regularly be updated and ensure that the system does not go down during popular times/business hours.

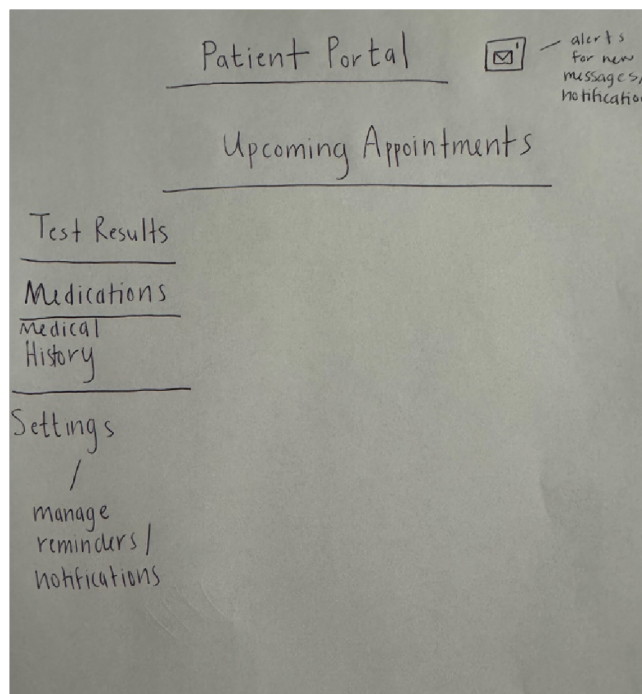
- User Interface Requirements:

No.	Priority	Description
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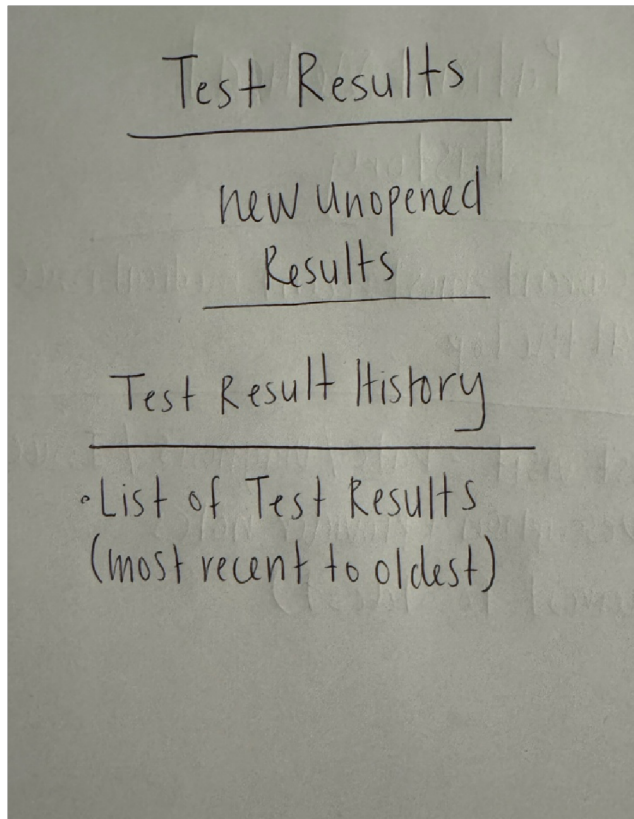
UI REQ-1	High	Patient portal home screen that shows appointments and alerts for new test results, messages from providers, updates to prescriptions, etc. Different tabs displayed help patients access full medical information, test results, etc.
UI REQ-2	High	Patients can view test results. New test results that have not been viewed yet will be highlighted in some way.
UI REQ-3	High	Patients can view full medical records. Here they will find current treatment plans and past medical information. Their full health history is completely accessible starting with the newest information at the top.
UI REQ-4	High	Administrators and medical staff (doctors, nurses, etc.)

		can see a staff view of the portal. There is a calendar to view all upcoming appointments. They are also able to access a list of current patients and navigate to these users. Here they can update records, send new prescriptions and refills, and communicate with patients if needed.
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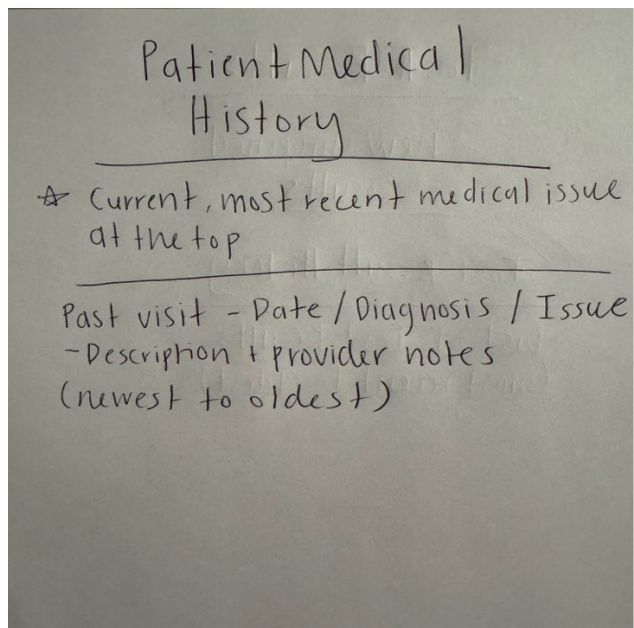
Patient Portal Home Screen Sketch



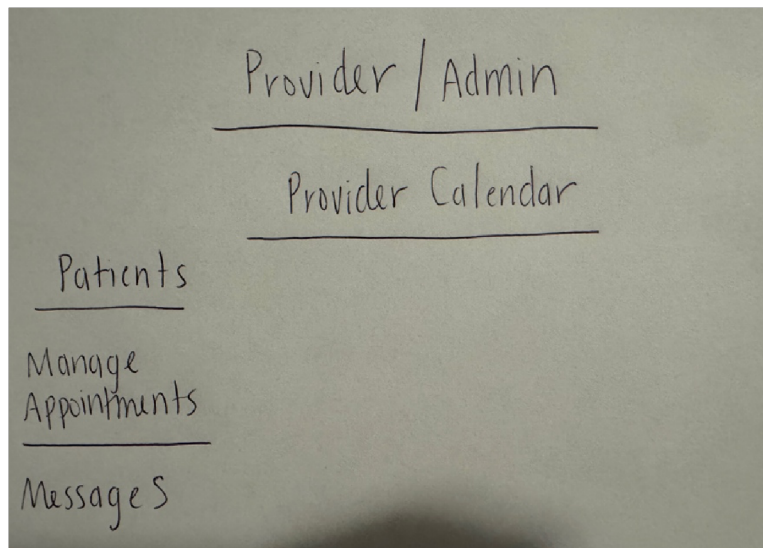
Test Result Page Sketch



Medical History Page Sketch



Provider/Admin Screen Sketch



Plan of Work:

Completed:

- Proposal
- Glossary of Terms
- System Requirements
- Sketched ideas of what the different screens in the portal will look like.

I have not started the actual coding yet, but I will begin on that this week. The work that I included in my timeline in my proposal will start to get checked off soon.

Functional Requirements Specification of Patient Tracker System

Stakeholders

- Patients
- Providers and Staff
- Admission Workers

Actors and Goals

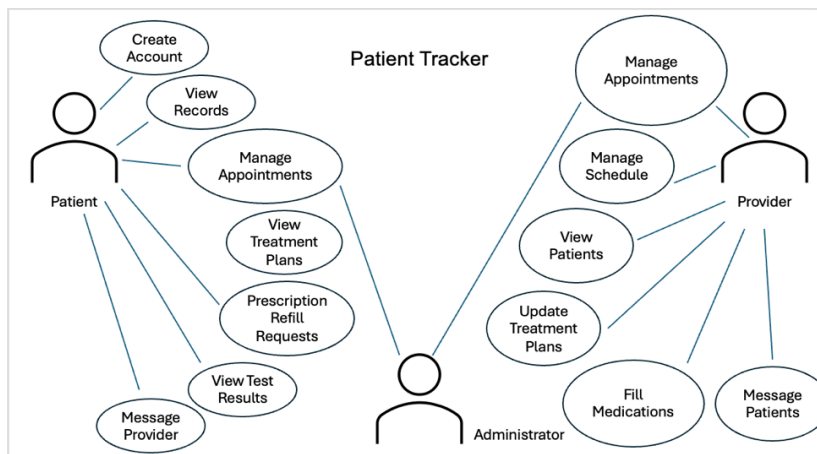
- Primary Actors
 - Patients can access all their medical records via the patient tracker system. They can set up an account without assistance or work. They can schedule appointments and request prescription refills from their doctors. Patients should be able to view their test results and even chat with their provider directly within the system.
 - Doctors and Staff can view their schedules and make changes as needed. They can see all their patients in the system. They can add to patients' treatment plans, fill prescriptions, and answer questions the patient asks.
- Secondary Actors
 - Administration Workers can go into the patient tracker system and manage appointments and add to the doctors' schedules. They can basically do everything for the patient via the system, but getting patients involved in their healthcare by giving them full access is the point of the patient tracker system.

Use Cases

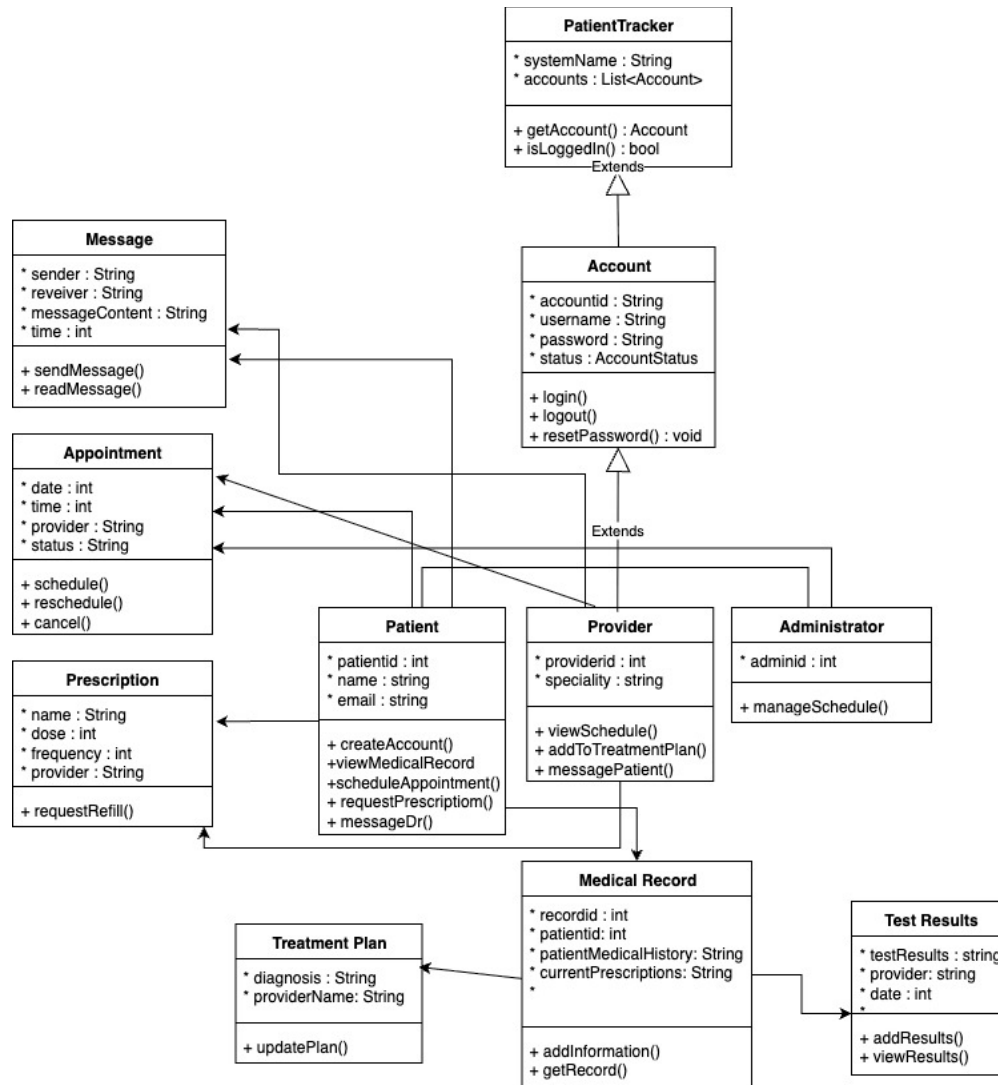
- Patient can create an account on their own (2).

- Patients can view full medical records (3).
- Patients can view their current treatment plans (2).
- Patients can schedule, reschedule, and cancel appointments in the system (3).
- Patients can request prescription refills (3).
- Patients can view their test results (3).
- Patients can talk with their providers (4).
- Providers can view their schedules (2).
- Providers can view patients (2).
- Providers can add to the patient treatment plans (3).
- Providers can fill prescriptions (3).
- Providers can communicate with their patients (4).
- Administrators can manage appointments (3).
- Administrators can schedule new appointments and add them to the provider schedules (3).
- Total: 40, which is about 20 engineer days

Case Diagram



Class Diagram



System Sequence Diagram and Activity Diagram

Patient Tracker System

System Sequence Diagrams

I chose “Schedule an Appointment” and “Request a Refill for a Prescription” to show sequence diagram examples for the Patient Tracker System.

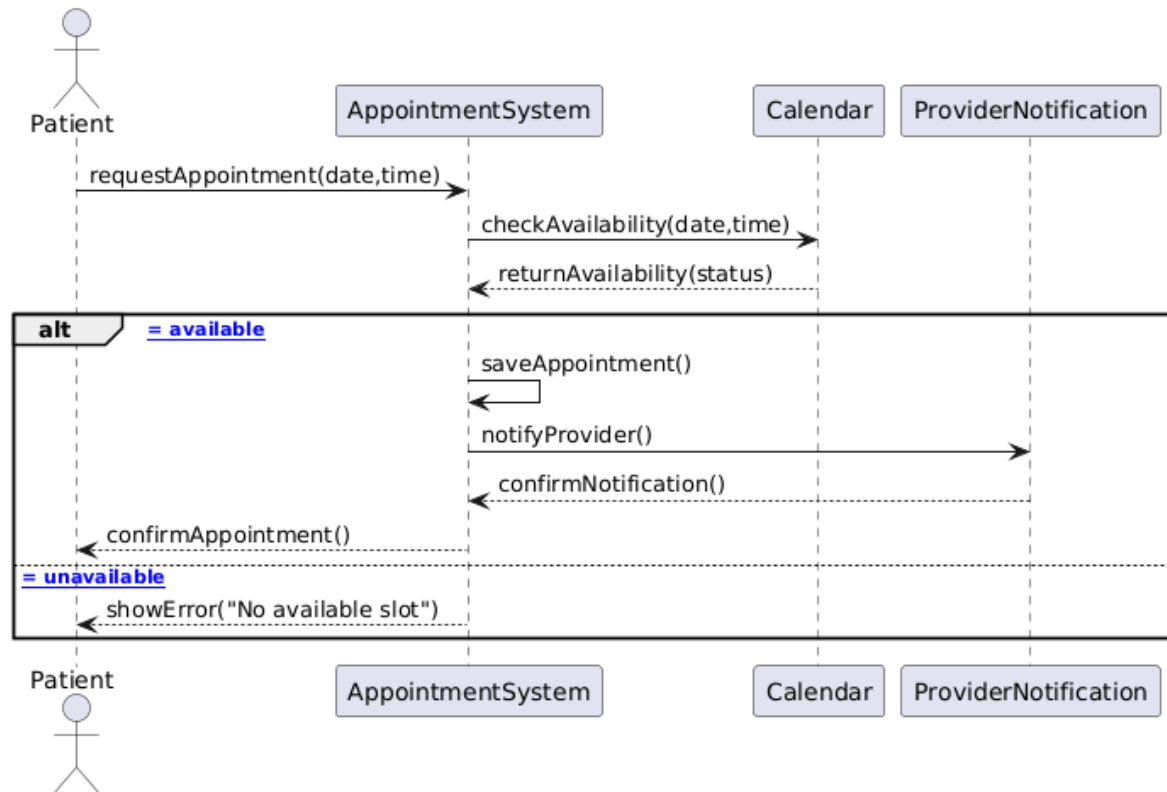
Schedule an Appointment:

Actor: Patient

Object: AppointmentSystem, Calendar, ProviderNotification

Steps of this interaction:

1. The patient requests a date and time for their appointment.
2. The patient tracker system checks the provider’s availability in the Calendar.
3. The calendar returns whether that appointment is available.
4. The system approves the appointment and stores it.
5. The system then notifies the provider of the appointment.
6. The patient gets confirmation from the system when the provider accepts the appointment.



Request a Refill for a Prescription:

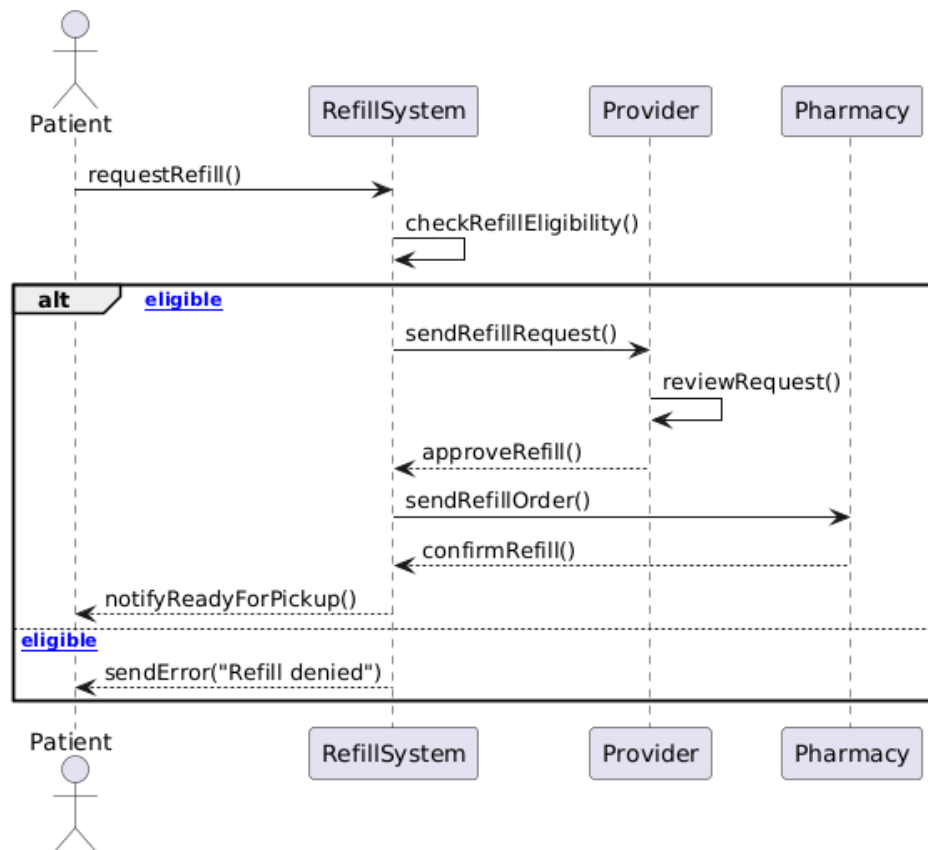
Actor: Patient

Object: RefillSystem, Provider, Pharmacy

Steps of this interaction:

1. The patient requests a refill for their prescription.
2. The RefillSystem will check whether the patient is due for a prescription refill.
3. If the patient is due for a refill, the RefillSystem will send the refill request to the Provider. If the prescription can't be refilled, the RefillSystem will send an error message back to the patient and deny the request.
4. The Provider will see the request and approve the refill.

5. The RefillSystem will send the request for a refill to the Pharmacy.
6. The Pharmacy will accept the request and refill the prescription.
7. The RefillSystem will notify the patient that the refill request has been sent to the Pharmacy and can be picked up there.



Activity Diagrams

Schedule an Appointment:

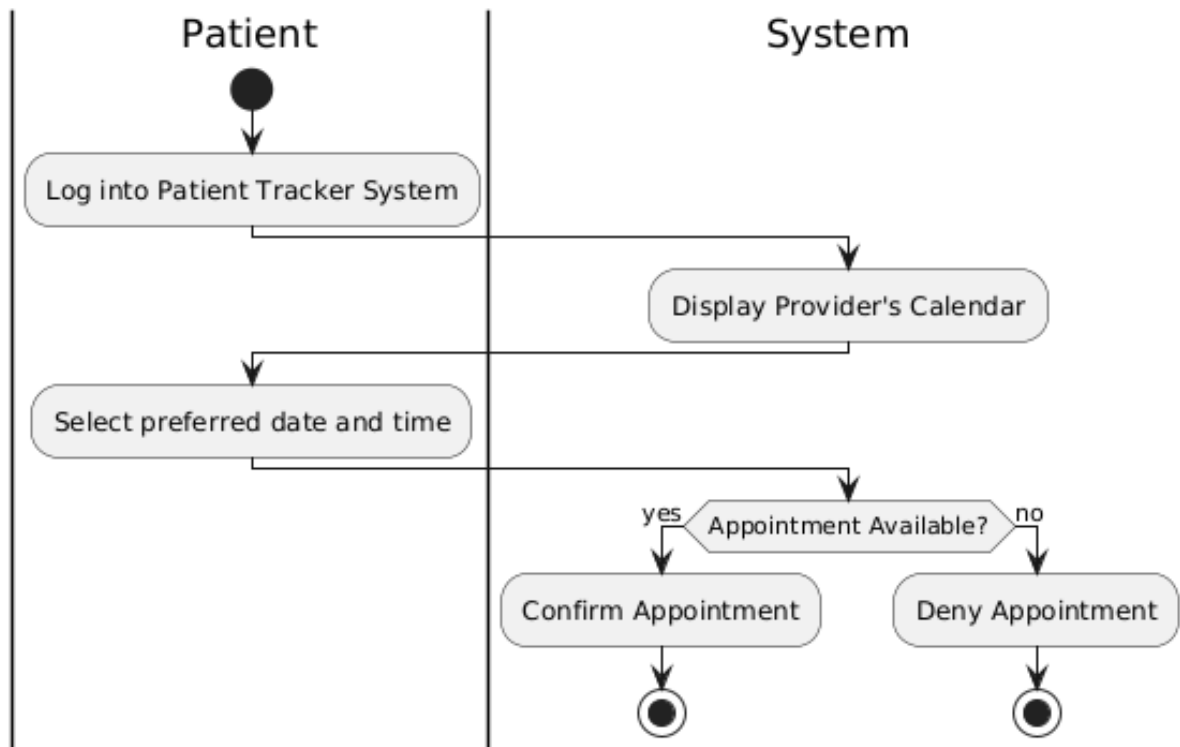
States:

- Initial State: The patient logs onto the patient tracker system.
 - Final State:
1. The appointment is accepted.

2. The appointment is denied because it isn't available.

Actions:

The patient pulls up the calendar to view the provider's availability. The system brings up the provider's calendar, and the patient can choose a day and time that works for them.



Request a Refill for a Prescription:

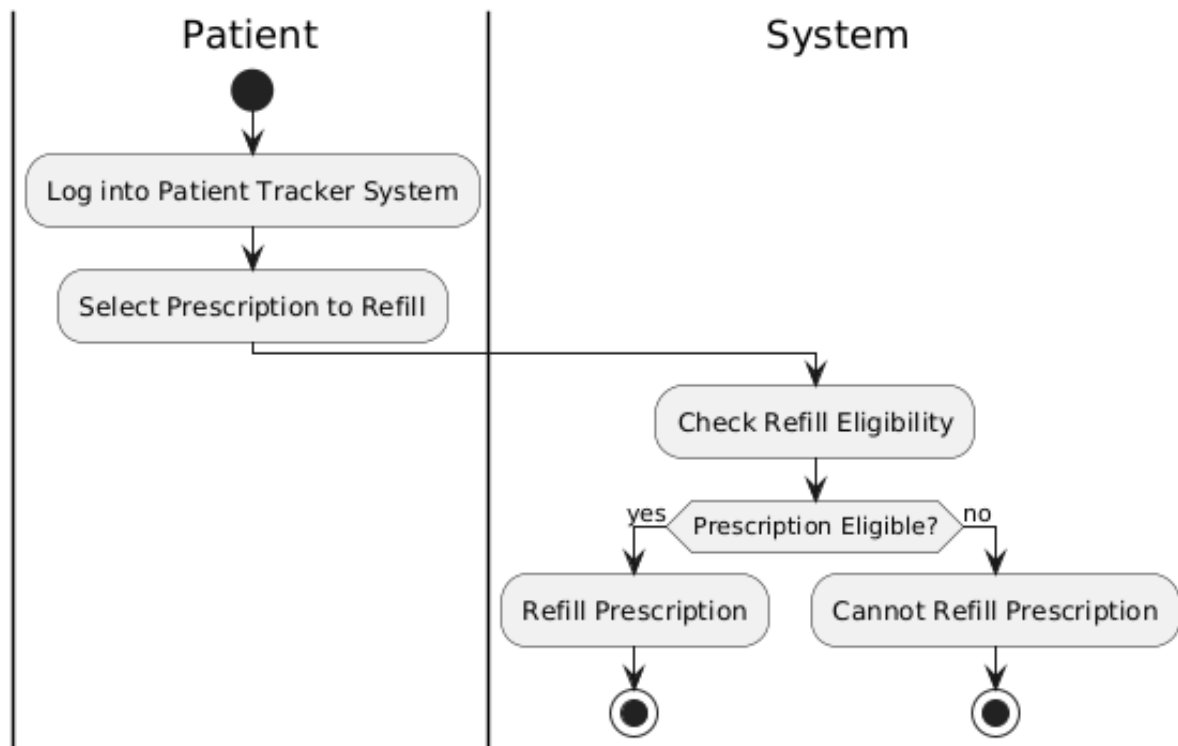
States

- Initial State: The patient logs onto the patient tracker system.
 - Final State:
1. The prescription is refilled.

2. The prescription is unable to be refilled.

Actions:

The patient selects the prescription they would like refilled. The system then checks whether the prescription is eligible to be refilled or not.



Project Planning and Development Approach:

Software

- **Frontend:** React.js • **Backend:** Python
- **Database:** MySQL

Hardware

- Computers
- A server

Network

- Quick internet access will be ideal, so patients can get the most up-to-date information about their health quickly.
- The network will be protected by firewalls/encryption, so that patient health information is not breached or shared with people who should not have access.

Development Plan:

- **Weeks 1–2:** The database will be set up, and the backend will be connected to the frontend.
- **Weeks 3–4:** User registration will be implemented, and login for different users (patients, doctors, admins, etc.) will be tested.
- **Weeks 5–6:** Patient account management will be set up so they can update information as it changes.

- **Weeks 7–8:** The appointment scheduling and the reminder system will be created.
- **Weeks 9–10:** Administrator management capabilities will be added, allowing them to access patient and staff accounts.
- **Weeks 11–12:** The system will be tested, and any errors will be fixed.
- **Weeks 13–14:** Usability and user experience will be improved.
- **Week 15:** Final testing will be performed, and the system will be prepared for release.